

Taylor N Perry

CSCI 496 Senior Portfolio

Spring 2019

In Partial Fulfillment of the
requirements for the degree of
Bachelor of Science

TAYLOR PERRY

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My positive attitude is what drives me to be the best in any environment the tech world has to offer. My keen eye for detail and willingness to ask questions helps me gain knowledge I need to not only do tasks that are given, but help those when they are having trouble. My optimistic attitude offers a different outlook on many hard-to-solve tasks and keeps me driven to always do my best.

EXPERIENCE

SEPTEMBER 2018 – CURRENT

BACKROOM LOGISTICS, TARGET

Keeping track of merchandise, Completes tasks before designated deadline, multitasking and time management, Excellent customer service helping to find desired items, use of technology to move/locate items, use of machine operated equipment, Understanding of logistics, Ability to talk to others and give instructions.

SUMMER 2009 – SUMMER 2018

SUMMER CAMP COUNSELOR, BSA COASTAL CAROLINA COUNCIL

First four years were as a merit badge counselor, ability to guide and teach boys of various ages from 13-17. Time management, Teaching skills to others and making sure they understand, Customer service, Going above and beyond on any task. Last 3 years I was Program director, in charge of a 60+ staff each year, schedule making/keeping, communications skills, training skills, presentation skills, running a camp with on average 200-500 campers per week for 5 weeks, organization and record keeping.

EDUCATION

DECEMBER 2019

COMPUTER SCIENCE B.S, CHARLESTON SOUTHERN UNIVERSITY

Familiar in, C++, Microsoft essentials(Word, Excel, PowerPoint, Teams, OneNote, Office, Visual Studio Code), Windows 7/8/8.1/10, Networking, Webpage design, Database management, System Analysis.

SKILLS

- Optimistic attitude
- Communications
- Willingness to take on any task
- Able to see my shortcomings and fix them
- Understanding and listening skills

ACTIVITIES

Eagle Scout/Vigil honor, Volunteer to Boy Scouts of America, Volunteer with elementary FIRST Lego club, Play trombone for local church annual Christmas concert, Student Government Association, Cyber Security Club(President), Member of Association for Computing Machinery(ACM) .

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Pg. 6 System Analysis: CpcApplication(Code Checker)

- Taylor Perry, Gray Team
- Creation date Fall 2018
- CSCI 495 System Analysis and Design
- Grade: 100/100
- Comments: N/a
-

Pg. 11 Hang Man

- CSCI 235 Procedural Programming
- You earned a 93.5 on this project and turned it in on 4/22/2017.

Pg. 26 German Credit(Data Mining)

- CSCI 442 Data Mining
- Grade N/A
- Comments: N/a

Pg. 33 16/32 Bit Extender Presentation

- CSCI 330 Computer Architecture
- 1/1 point
- Comments : N/a

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CPCApplication source Code

```

package cpcapplication;

/*
 * @author Taylor Perry
 */

import java.io.*;
import java.io.File;
import java.io.File.*;
import java.io.IOException;
import java.nio.file.CopyOption;
import java.nio.file.Files;
import java.nio.file.Paths;
import java.nio.file.StandardCopyOption;


public class Library {

    public static void main(String[] args) throws IOException{

    }

    Library() throws IOException { //constructor

        // gets the username of the current user
        String username = System.getProperty("user.name");

        //creates a file instance of the library and its path
        File file=new File("C:\\Users\\" + username + "\\Documents\\cpcapplication\\library");

        boolean exists = file.isDirectory();

        //if the library directory does not exist then create it else do nothing
        if (!exists){

            System.out.println("The Library does not exist");
        }
    }
}

```

```

file.getAbsolutePath().mkdirs();
if (exists) {
    System.out.println("Library was created!");
} else {
    System.out.println("Failed to create the Library!");
}
} else {
    System.out.println("The Library exists");
}
}

```

```

static void addAFile(File src) throws FileNotFoundException, IOException {

    //checks to see if the file that is being added to the Library directory exist
    if (src.exists()){

        //gets the username of the current user
        String username = System.getProperty("user.name");

        //creates a file instance of the destination file
        File dest=new File("C:\\Users\\" + username +"\\Documents\\cpcapplication\\library\\" +
src.getName());

        //creates the file into the Library directory
        dest.createNewFile();

        //creates the options used for the copy
        CopyOption[] options = new CopyOption[]{

```

```

        StandardCopyOption.REPLACE_EXISTING,
        StandardCopyOption.COPY_ATTRIBUTES
    };

    //copy the contents of the file to the destination file in the Library directory
    Files.copy(src.toPath(), dest.toPath(), options);

    System.out.println("File: " + dest.getName() + " has been copied to the Library");
} else
{
    System.out.println("File does not exist or path is bad");
}
}

static void remAFile(File file){
    //gets the username of the current user
    String username = System.getProperty("user.name");

    //creates a file instance of the destination file
    File libFile = new File("C:\\Users\\" + username + "\\Documents\\cpcapplication\\library\\" +
file.getName())

    //checks to see if the file exists in the Library
    if (libFile.exists())
    {
        libFile.delete();

        System.out.println("File in library with the name of " + file.getName() + " has been deleted from the
library");
    }
else {
        System.out.println("There was no file with the name of " + file.getName() + " in the library")
    }
}
}

```

CPC Application

Comparisons

[Go To Database Page](#)

Choose File 1 Below to Upload

[test.txt](#)

☒ Save to the Database

Choose File 2 Below to Upload

[test 2.txt](#)

☐ Save to the Database

[Compare File 1 to File 2](#)

===== OR =====

Plagiarism Checker

Paste Code Below for File 1 Comparison

☐ Save to the Database

[Compare File 1 to Pasted Code](#)

CPCApplication Screenshot 1

CPC Application

Results

Raw Code Comparison

File 1 Code	File 2 Code
CPCApplication test	This is a test

Percentage Report: 0

Make Another Comparison

CPCApplication Screenshot 2

CPC Application

Database

Go To Comparisons Page

Choose File to Compare Below	Paste Text Below for Database Compar...
Browse...	
Choose Database Interaction:	
Add To The Database	Name Your File Before Adding to The ...
Remove From The Database	
	Add Paste Contents to Database

CPC Application Screenshot 3

Hang Man

User is to input a letter and guess the phrase that comes from a .txt file.

The game will have you choose a difficulty, the easier it is the less unique letters it will have. It shows the phrase as ' _ ' and displays incorrectly guessed letters and will display that the user has won or lost based on if they won or lost. then ask the user if they wish to play again.

By Taylor Perry.

```
*/
```

```
#include<iostream>
```

```
#include<iomanip>
```

```
#include<fstream>
```

```
#include<cctype>
```

```
#include<string>
```

```
#include<ctime>
```

```
#include<cstring>
```

```
using namespace std;
```

```
struct Phrase
```

```
{
```

```
    string text;
```

```
    string::size_type guessRequired;
```

```
    bool isUsed = false;
```

```
};
```

```
//can handle up to 100 phrases from a text file. called phrases.txt
```

```
const int ARRAY_LENGTH = 100;
```

```
//So that there wont be any duplicate answers this counts how many of each type
```

```
//of letter is in the phrase
```

```
int uniqueLetterCount (string& word);
```

```

//Opens & closes file with all the phrases in it.
int loadPhrasesFromFile(string fileName, Phrase array[], int arrayLength);

//aranges phrases
void sortPhrases(Phrase array[], int arrayLength);

//put phrases in order from least to greatest unique letters
void swap(Phrase array[], int num1, int num2);

//prints the phrase for user to see
void printPhrases(Phrase array[], int arrayLength);

//Lets user choose the difficulty of the phrases to come.
int getDifficultyLevel();

//based on difficulty choice will choose a random phrase in said difficulty.
int choosePhraseBasedDifficulty(int difficulty, int phraseCount,
                                Phrase phrases[]);

void runGame(string& phrase, int guessRequired);

//This calls the outline of the hangman platform.
void drawGallows(int missCount);


int main ()
{
    //gets the number of phrases from .txt
    int phraseCount;

    //allows a random number to pick out a different phrase
    unsigned int seed = static_cast<unsigned int> (time(nullptr));

    //    const int ARRAY_LENGTH = 100;
    Phrase phrases[ARRAY_LENGTH];

    //allows asks the user if they want to play again.
    bool keepPlaying = false;

    int phraseIndex;

    char letter;

```

```

//phrases count is equal to number of files from .txt file.
phraseCount = loadPhrasesFromFile("phrases.txt", phrases, ARRAY_LENGTH);
//function sorts phrases.
sortPhrases(phrases, phraseCount);

//loop allows you to run the program as many times as you like.
do
{
    phraseIndex = choosePhraseBasedDifficulty(getDifficultyLevel(),
                                              phraseCount, phrases);

    phrases[phraseIndex].isUsed = true;
    //calls the function run game which is what you see on screen
    runGame(phrases[phraseIndex].text, phrases[phraseIndex].guessRequired);
    //asks if you want to play again.
    cout << "Do you wish to play again? (y/n) " << endl;
    //gets answer to if you want to play again.
    cin >> letter;

    if (letter == 'y')
        keepPlaying = true;
    else
        keepPlaying = false;

    } while(keepPlaying);

return 0;
}

//function used to figure out how many different letters are used in phrases.

```

```

int uniqueLetterCount (string& word)
{
    string uniqueLetters = "";
    char letter;
    int length = word.length();

    //letters will be all lowercase. if the letter is between a-z then it will
    //find that letter and save it as a unique letter untill there are no more
    //letters in the words length.
    for (int index = 0; index < length; ++index)
    {
        letter = tolower(word[index]);
        if(letter >= 'a' && letter <= 'z' && uniqueLetters.find(letter)
            == string::npos)
            uniqueLetters += letter;
    }
    //return value is how many letters are going to be used in the phrase
    //(no repeats)
    return uniqueLetters.length();
}

//gets the phrases from .txt file.
int loadPhrasesFromFile(string fileName, Phrase array[], int arrayLength)
{
    string phrase = "";
    ifstream fileIn;
    int index;

    //opens the file with phrases.
    fileIn.open(fileName);

    //this will get the number of phrases in the .txt untill it gets to the

```

```

//end of the file.
for(index = 0; index < arrayLength && !fileIn.eof(); ++index)
{
    getline(fileIn, phrase);

    //declaring values in the struct.
    array[index].text = phrase;
    array[index].guessRequired = uniqueLetterCount(phrase);
    array[index].isUsed = false;
}
fileIn.close();

return index;
}

//function used to help with sortPhrases function, algorithm to move phrases.
void swap(Phrase array[], int num1, int num2)
{
    Phrase num3;
    num3 = array[num1];
    array[num1] = array[num2];
    array[num2] = num3;
}

//Function uses the swap function to move phrases with least unique letters
//to the top and phrases with the most unique to the bot.
void sortPhrases(Phrase array[], int arrayLength)
{
    //the least amount of unique letters cannot go below 0.
    int smallestIndex = 0;

    for (int index1 = 0; index1 < arrayLength; ++index1)

```

```

{
    for (int index2 = index1; index2 < arrayLength; ++index2)
    {
        if (array[smallestIndex].guessRequired >
            array[index2].guessRequired)
        {
            smallestIndex = index2;
        }
    }

    swap(array, smallestIndex, index1);
}
}

```

//Function takes the random phrase and turns all letters into '_'.

string phraseWithBlanks(const string& text, const string& guessedLetters)

```

{
    //any single character is seen as a lowercase letter.
    char lower;
    //your future string will have '_' for each letter in a phrase.
    string blankPhrase;
    //finds out how many letters are in the phrase and replaces the character
    //with a '_'.
    for(string::size_type index = 0; index < text.length(); ++index)
    {
        lower = tolower(text[index]);
        if(lower >= 'a' && lower <= 'z' && guessedLetters.find(lower)
            == string::npos || lower >= 'A' && lower <= 'Z' &&
            guessedLetters.find(text[index]) == string::npos)
        {

```

```

        blankPhrase += "_ ";
    }
    else//if its not a upper or lower case letter it will just leave the
        //space as is.
    {
        blankPhrase += text[index];
        blankPhrase += " ";
    }
}
return blankPhrase;//returns either a '_' or " ".
}

```

//where you can show off your hangman skills.. or practice..

```

int getDifficultyLevel()
{
    //difficulty is a whole number (1,2, or 3) therefore it's an int.
    int difficulty;

    //Displays the option from the user to mae a choice of difficulty.
    cout << "Difficulty levels: " << endl;
    cout << "    1    Easy"<< endl;
    cout << "    2    Medium" << endl;
    cout << "    3    Hard" << endl;
    cout << "Pick a difficulty level (1-3): " << endl;

    //what ever unmbner you enter will be called the difficulty.
    cin >> difficulty;

    //if the cin is not 1, 2, or 3, then it will output a error message.
    while (!cin || (difficulty < 1 || difficulty > 3))
    {
        cout << "Invalid letter, please enter a difficulty 1-3" << endl;
    }
}

```

```

    cin.clear();

    //Enter as many numbers you want but its only going to look at the first
    //one.
    cin.ignore(INT_MAX, '\n');
    //Try entering in the right cin input(hopefully)
    cin >> difficulty;
    cout << endl;
}

//depending oh the cin difficulty you entered in it will display the
//difffficulty at the start of the game.
if(difficulty == 1)
{
    cout << "*Easy*" << endl;
}
else if(difficulty == 2)
{
    cout << "*Medium*" << endl;
}
else
{
    cout << "*Hard*" << endl;
}

return difficulty;
}

//Function puts list of phrases into 3 groups based on total number of phrases
// then divides into 3(based on the difficulty)
int choosePhraseBasedDifficulty(int difficulty, int phraseCount,
                                Phrase phrases[])
{
    int index = 0;

```

```

int range = phraseCount / 3;
int min = range * (difficulty - 1);

if(difficulty == 3)
{
    range += phraseCount % 3;
}
do
{
    index = rand() % range + min;

    //if you continue to play the game over and over will look for a phrase
    // where isUsed is false.
    while(phrases[index].isUsed);

    return index;
}

//this function is the game. Displays what we see and takes in all the
//inputs to play the game.
void runGame(string& phrase, int guessRequired)
{
    //start the game with 0 missed letters
    int missCount = 0;

    //All letters you input are put into a string to be sorted later.
    string guessedLetters = "";

    //this is the individual letter that you input.
    char guessedLetter;

    //start game with 0 guesses, untill you input first character.
    int numOfGuessedLetters = 0;

    //keeps track of the guessed letters you got correct.
    string badGuess = "";

```

```

//The space between each pictures so the previous screen wont be seen.
cout << string(18, '\n');

//calls the function that displays the gallows platform.
drawGallows(missCount);
//shows the incorrect guesses so you don't guess them again.
cout << "Incorrect guesses: " << badGuess << endl;
//shows the letters of the choosen phrase as '_ '.
cout << phraseWithBlanks(phrase, guessedLetters) << endl;

do
{
    //loops the question of asking for a letter.
    cout << "Guess a Letter: ";
    //that current letter is put into the string of guessedletters and
    //will then be put into a correct/bad guess.
    cin >> guessedLetter;
    //no matter what was entered as the guessedLetter it will be seen as
    //a lower case number.
    guessedLetter = tolower(guessedLetter);

    cout << string(18, '\n');

    //If cin guessedLetter is not a letter program outputs not valid.
    if(guessedLetter < 'a' || guessedLetter > 'z')
    {
        cout << guessedLetter << " is not a valid guess!!!" << endl;
    }
    //Or it will let user know that it has already used the same letter
    //twice.
    else if(guessedLetters.find(guessedLetter) != string::npos)

```

```

        cout << "-ERROR- You have already guessed " << guessedLetter
            << endl;

//even if the letter is uppercase in the phrase it will recognize it.
else if(phrase.find(guessedLetter) != string::npos ||
        phrase.find(toupper(guessedLetter)) != string::npos)
{
    //when you get a letter right it will congratulate you.
    cout << "Good Guess Kid-o!" << endl;

    //adds 1 to unmbner of guessed letters so once you get all the
    //guessed letters for the phrase you win.
    ++numOfGuessedLetters;
    guessedLetters += guessedLetter;
}
else
{
    //if you guess the incorrect letter
    cout << "Sorry, That was a Incorrect guess!" << endl;

    //this will count as a bad guess and will add to the miss counter.
    badGuess += guessedLetter;
    badGuess += " ";
    guessedLetters += guessedLetter;
    ++missCount;
}

//updates the gallows to the correct case if the user got a badGuess.
drawGallows(missCount);

cout << "Incorrect guesses: " << badGuess << endl;
cout << phraseWithBlanks(phrase, guessedLetters) << endl;
}

//if you have less then 5 missed guesses and you didnt go over the Required
//letter counn you win.

```

```
while (missCount < 5 && numOfGuessedLetters < guessRequired);
```

```
if(missCount != 5)
```

```
{
```

```
    cout << endl;
```

```
    cout << "YOU WON!!!" << endl;
```

```
    cout << endl;
```

```
}
```

```
else
```

```
//if you di get 5 badguesses and go over required letters needed you lose.
```

```
{
```

```
    cout << endl;
```

```
    cout << "YOU LOST!!!" << endl;
```

```
    cout << endl;
```

```
}
```

```
//shows the phrase at the end.
```

```
cout << "The phrase was \" << phrase << "\" << endl;
```

```
}
```

```
//funciton provided by the very nice, intelligent, hard working teacher.
```

```
void drawGallows(int missCount)
```

```
{
```

```
    // Output the top of the gallows
```

```
    cout
```

```
        << "  +-----+  \n"
```

```
        << "  |   |   \n";
```

```
    // Output the stand and body
```

```
    switch (missCount)
```

```
    {
```

```
        case 0:
```

cout

```
<< " | \n"
<< " | \n"
<< " | \n"
<< " | \n";
```

break;

case 1:

cout

```
<< " | O \n"
<< " | | \n"
<< " | \n"
<< " | \n";
```

break;

case 2:

cout

```
<< " | O \n"
<< " | /| \n"
<< " | \n"
<< " | \n";
```

break;

case 3:

// The "\\ ' will output as '\ ', because it is a special char

cout

```
<< " | O \n"
<< " | /|\\ \n"
<< " | \n"
<< " | \n";
```

break;

case 4:

cout

```
<< " | O \n"
```

```
<< " | /|\ \n"
```

```
<< " | \ \n"
```

```
<< " | \n";
```

break;

default:

cout

```
<< " | O \n"
```

```
<< " | /|\ \n"
```

```
<< " | / \ \n"
```

```
<< " |You're Dead\n";
```

}

// Output the base

```
cout << " =====\n" << endl;
```

}

```
Choose a difficulty:
(1) Easy
(2) Medium
(3) Hard
Pick a difficulty level (1 to 3):
```

Hangman Screenshot 1

Data Mining: German Credit

```

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.cross_validation import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix

#change data it was wat you call to call your graph
data = pd.read_csv('german_credit.csv', sep=',')

#prints the data set
print(data)

#this will generate the histogram
for column in data:
    plt.hist(data.loc[:,column].values)
    plt.title('Histogram for' + column + 'column of the German Credit dataset')
    plt.xlabel('Value in column' + column + 'column' )
    plt.ylabel('Number of Occurances')
    plt.show()

#change freq to mean something else?
freq = data[column].value_counts()
labels = data[column].unique()
num_val = len(labels)

#gernerate a bar chart based on the frequency each value in a column occurs.
plt.title('Bar chart for' + column + 'Column')
plt.bar(labels, freq, align = 'center')

```

```
plt.xticks(labels)
```

```
plt.show()
```

```
#generate a pie chart based on the frequency each value in a column occurs
```

```
plt.title('Pie Chart for' + column + 'column')
```

```
plt.pie(freq, labels=labels)
```

```
plt.show()
```

```
#generate a box plot for each column
```

```
plt.title('Box Plot for' + column + 'column')
```

```
plt.boxplot(data.loc[:,column].values)
```

```
plt.ylabel(column)
```

```
plt.show()
```

```
#enables the split of the data to do the training sets
```

```
#import numpy to import data
```

```
import numpy as np
```

```
#imports the csv file and skips the first row
```

```
data = np.loadtxt('german_credit.csv', skiprows = 1, delimiter = ',')
```

```
#give data of class information as the name of each column : gets all of the row, 0 sets it to the first column
```

```
#chance ClassDets just in case
```

```
ClassDets = data[:, 0]
```

```
#give the data of the features of the matrix
```

```
Features = data[:, :-1]
```

```
#split the data into a training and testing set
```

```

FTraining, FTesting, CDTraining, CDTesting = train_test_split(Features, ClassDets, test_size = 0.3, random_state = 0)

#transforms the test and trainign data
scaler = StandardScaler()
scaler.fit(FTraining)

FTraining = scaler.transform(FTraining)
FTesting = scaler.transform(FTesting)

#create the model with 5 neighebers
classifier = KNeighborsClassifier(n_neighbors=5)

#fit the model on the training data
classifier.fit(FTraining, CDTraining)

predict = classifier.predict(FTesting)

#prints the confusion matrix for the prediction based on the test data
print(confusion_matrix(CDTesting, predict))

# import svm model
from sklearn import svm

#Create a svm Classifier
clf = svm.SVC(kernel='linear') #Linear Kernal

#train the model using training sets
clf.fit(FTraining, CDTraining)

#predict the responce for test dataset
predict = clf.predict(FTesting)

```

```
#prints the confusion matrix for the prediction based on the test data
```

```
print(confusion_matrix(CDTesting, predict))
```

```
#uses the training sets to generate a model
```

```
MLP = MLPClassifier()
```

```
#uses the training sets to generate a model
```

```
fit= MLP.fit(FTraining, CDTraining)
```

```
#creates a prediction based on the fit above
```

```
#predicted change to name
```

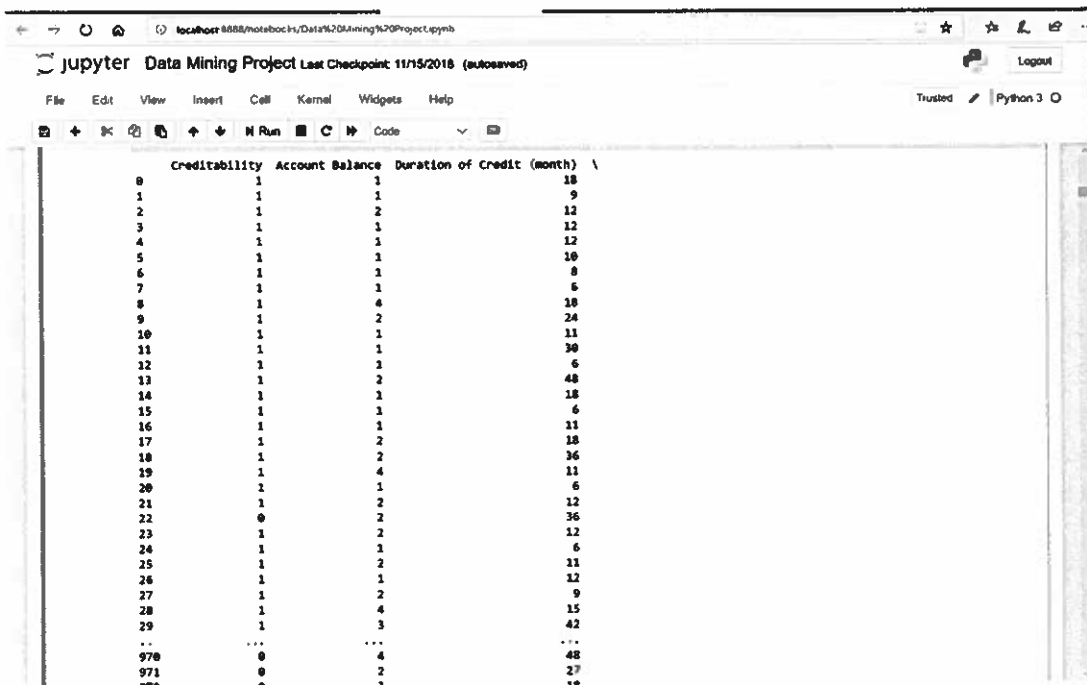
```
predicted= fit.predict(FTesting)
```

```
#generates a confusion matrix based on the real dat versus the predicted data
```

```
result = confusion_matrix(CDTesting, predicted)
```

```
#prints the final confusion matrix
```

```
print(result)
```



	Creditability	Account Balance	Duration of Credit (month)	
0	1	1	18	
1	1	1	9	
2	1	2	12	
3	1	1	12	
4	1	1	12	
5	1	1	10	
6	1	1	8	
7	1	1	6	
8	1	4	18	
9	1	2	24	
10	1	1	11	
11	1	1	30	
12	1	1	6	
13	1	2	48	
14	1	1	18	
15	1	1	6	
16	1	1	11	
17	1	2	18	
18	1	2	36	
19	1	4	11	
20	1	1	6	
21	1	2	12	
22	0	2	36	
23	1	2	12	
24	1	1	6	
25	1	2	11	
26	1	1	12	
27	1	2	9	
28	1	4	15	
29	1	3	42	
...	...	...	...	
970	0	4	48	
971	0	2	27	
...	...	...	...	

Jupyter Data Mining Project Last Checkpoint: 11/15/2018 (autosaved)



Logout

File Edit View Insert Cell Kernel Widgets Help

Trusted Python 3

Run Code

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29      4      56      3
..      ...      ...      ...
970     3      33      3
971     4      42      2
972     1      37      3
973     3      29      3
974     2      23      3
975     4      34      1
976     2      32      1
977     4      68      1
978     2      25      3
979     4      35      3
980     2      22      3
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982     3      30      3
983     3      25      1
984     1      34      3
985     3      42      2
986     2      45      3
987     2      20      3
988     3      23      3
989     2      20      3
990     1      27      3
991     4      44      3
992     2      51      3
993     1      33      1
994     2      28      3
995     1      21      3
996     1      45      3
997     4      30      3
998     4      52      3
999     2      31      3

```

Type of apartment No of Credits at this Bank Occupation \

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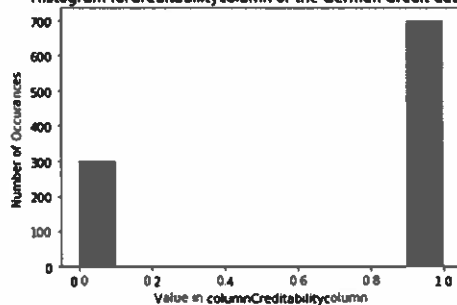
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Trusted Python 3

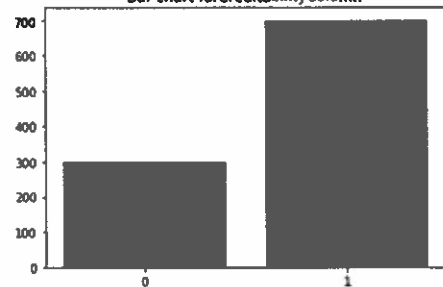
Run Code

[1000 rows x 21 columns]

Histogram for Creditability column of the German Credit dataset



Bar chart for Creditability Column





Logout

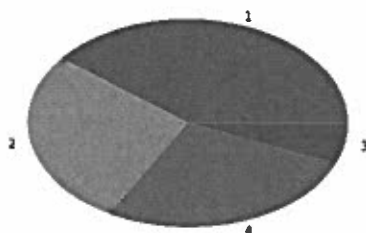
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Trusted Python 3

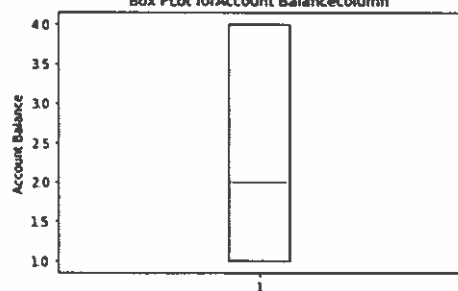
Run Code



Pie Chart for Account Balance column



Box Plot for Account Balance column



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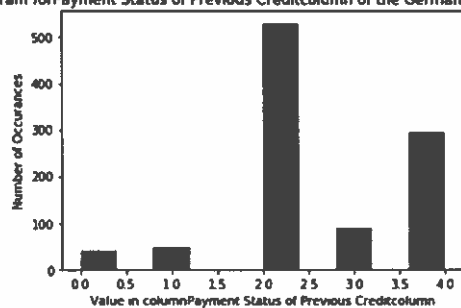
Trusted Python 3

Run Code

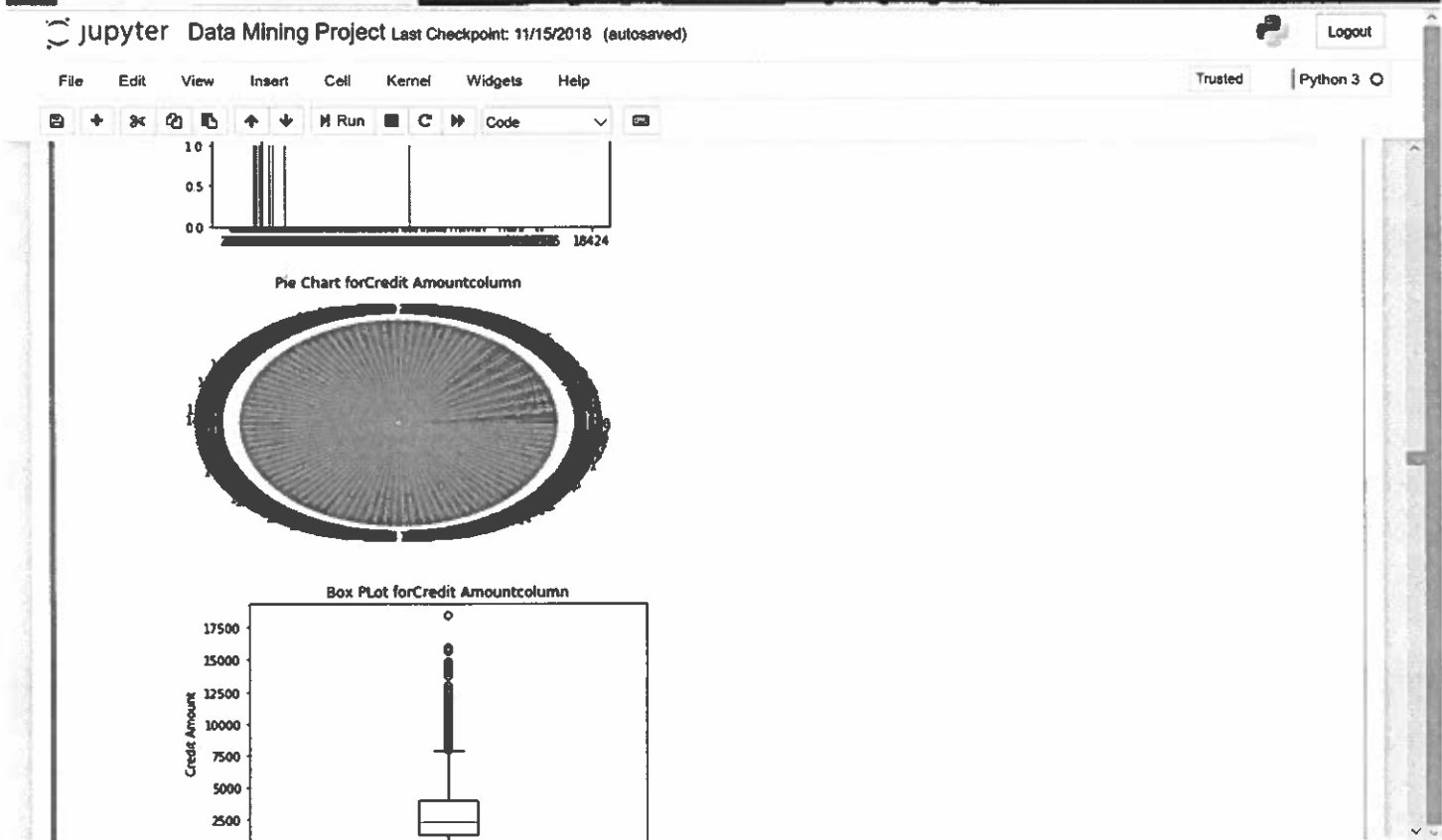
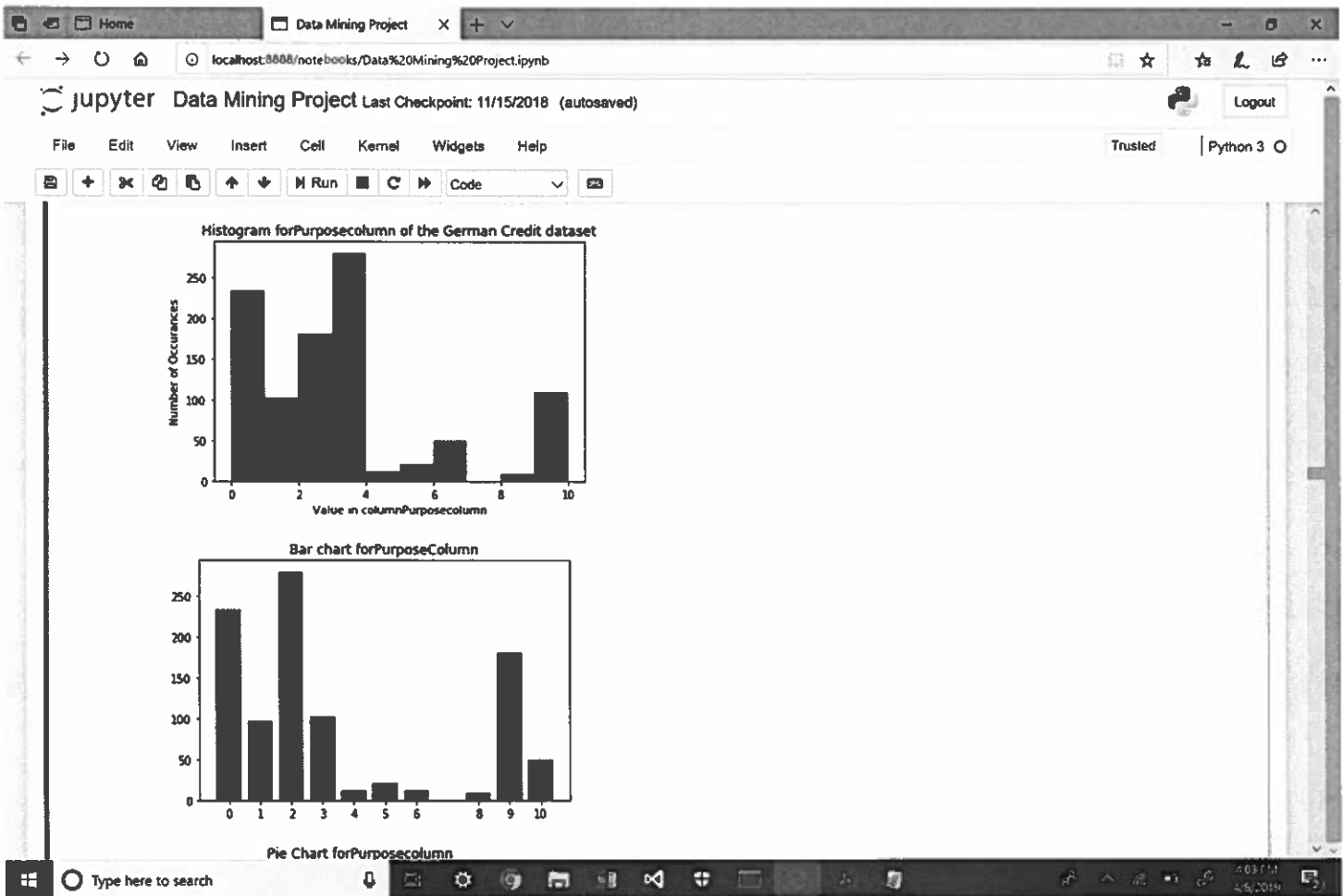
Box Plot for Duration of Credit (month) column



Histogram for Payment Status of Previous Credit column of the German Credit dataset



Bar chart for Payment Status of Previous Credit column



Computer Architecture: Sign Extender

SIGN EXTENDER

BY TAYLOR PERRY(PYROCAMPER)

WHAT IS IT??

- The process of increasing bits.
- Most commonly returns a value of 32 bits.
- If the first value is a 0, then it will add on more 0's keeping it positive
- If the first value is 1, then it will add 1's to keep the number negative.

WHY DO WE NEED A SIGN EXTENDER

- The memory will not always give out 32 bit integers
- Done before you add/subtract 2 signed integers
 - Ex: add two signed integers but one value is 16 bits and the other is 32



TEST CODE

```

1  void sign_extender_test()
2
3  reg_t[15:0] user_registers_a 16 bit number
4
5  wire [31:0] output_wire // here what we call the output
6
7  sign_extender sign_extender(sign_extender_in);
8
9  initial
10 begin
11   // user = 16'0000000000000000 // give it a register and
12   #20
13   $display("Test 1: 16'0000000000000000");
14
15   // user = 16'0000000000000001 // give it a register and
16   #20
17   $display("Test 2: 16'0000000000000001");
18
19   // user = 16'0000000000000010 // still using if it looks at the first 0
20   #20
21   $display("Test 3: 16'0000000000000010");
22
23   // user = 16'0000000000000011 // see what happens if we give it a 12 bit
24   #20
25   $display("Test 4: 16'0000000000000011");
26
27   // user = 16'0000000000000100 // give it a 6 bit
28   #20
29   $display("Test 5: 16'0000000000000100");
30
31   #40
32 end
33
34 // end of file
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PROBLEMS I HAD

- If you give it the wrong value(16 bits) it will output the wrong.
- This is structural flow
- The delay is 0
 - There are no gates therefor no gate delay

THE END

```
module extender(unextend, extended);  
input [15:0] unextend;  
output[31:0] extended;  
  
assign extended = { {16{unextend[15]}}, unextend };  
  
endmodule
```

//gate way is 0 because there are no gates and is done "instantly"

Ethics Papers:

1. Applied Networking Ethic Paper
-Pg. 37
2. Procedural Programing (CSCI 235)Ethics Paper
- Pg. 39
3. Data Structures ()Ethic Paper
-Pg. 42
4. Data Mining Review and Analysis
- Pg. 45

WHO'S CODE IS IT ANYWAYS?

The idea and thought of owning a piece of code is something not many people will understand and could probably not ever relate to. As much as it would be nice to have information out in the open, free to use, that just goes against the principal that nothing is free in this world. Either it was the time that a programmer put into a code, or paying to use a useful tool that is being used to help write this essay. Would it be beneficial to have these powerful and useful tools for free, oh course, but that will most likely never happen. But what draws the line in the sand on what is reasonable to call ours and not share with the world?

Now that we as a society have advanced and have examples in the past to compare and help better our future it should be no problem deciding if the code programmers develop should be open to the public or can be licensed off and if you have enough money you can afford it. Regardless that decision is up to the programmer himself, or is it. Some people want the world to grow faster and more advanced and the best way to do that is if a lot of people can work on something. Just like with electric cars, a certain company released its plans on how to make efficient electric cars because they wanted to save and advance this revolutionary movement that will help advance the world.

Even if you make this amazing code or application the organization you work for just might have a different policy regarding what you do. At Target where I work they have a policy that anything you do to better the store such as a design or product, Target owns the Rights to that idea and you have to give it over to them. It sounds harsh but from a tech company perspective if you provided the training to better your team member and they make this amazing display and leave to sell it for thousands, you just gave away free training and got nothing from it. So even if you make this amazing code the company that provided the tools and training have every right to what you made.

When electronic music and editing were introduced there was a new world of possibility for artists and DJ's. During this time of discovery artists would take old songs and remix them into a newer song but this new song, that included the song of another, would be a hit and the one who made the remix would make all the money. After a major trial we now have the copyright laws that exist today, so maybe to have a better understanding for laws on code that big trial will need to happen. Some good examples of some recent lawsuits over code and programs would be with the tech giants Samsung and Apple. With smartphones you can only change the software so much and eventually to get the best and fastest code its good to use a code that may be similar.

Even though you may have a breakthrough idea and would want to give it away to others for free but your company is against that, It sucks but its only fair to the company that they get something back in the person they invested time and money into. The good side of that is that some major company are giving away code for free or for a really reduced price, for example Microsoft gives away a free version of Office for a year if you are a student. This shows that if you are a big enough company you can give stuff away for free.

In conclusion finding a idea to side with is very situational, and newer versions of code and applications are popping up which makes it even harder to define what is legal and what is free. As much as it is nice to have free things, looking at it from business view it isn't smart to just give things away and expect nothing in return.

Procedural Programming Ethic Paper: "Morals of Ad Blocking"

Taylor Perry

CSCI 235

2/20/2017

Professor Hayes

Morals of Ad Blocking

The first ad was a pop-up was a nice side banner that appeared on dial-up computers back in 1994. I can see the frustration of trying to use the already slow internet and the first thing the computer loads is the ad that has nothing to do with what you were looking for. Companies spend a major percentage of their income on advertising and why? Do I really need to know that boots are going to be 30 percent off this weekend only or is it to reach as many people as possible? One thing is sure, online advertising is completely different then what it used to be.

In Luke 6:31 it states "Do to others as you would have them do to you", and I honestly base a lot of this idea to my life. I don't even feel bad overusing it because it's a simple yet effective way of creating a fairer world. Now applying this idea to online advertising I can't help but feel like online ads today have taken it too far. Ads went from passive to annoying eye sores that, in my mind, can make me totally ignore the page based on if it looks like my computer is going to get a virus. Even though I know ad-blocking hurts the creator of the website I still have no problem in using an ad-blocker.

With everything, there is a wrong way and a right way to something. Advertising is a great way of letting a vast amount of people know about your business. It's an even better deal for website owners, newspapers, and even Youtube artists to earn more money. With Youtube now changing its policy on ads you are no longer forced to watch a 30 second ad for a 3-minute video. This is good, this is Youtube seeing that advertising may be getting out of control and you can't just sit by and pretend everything is okay. Since ad-

blocking was introduced it did mean that the advertising company had to change how they did things or else they would just be wasting money on ads no one will see. So, they either had to tone down the ad or they took it a step further and didn't advertise the company, but rather bought your information. Facebook is a great example to this.

Lastly when talking about ads and ad-blocking I can't help but feel the creator is also responsible if ads get out of control. If an ad is out of control and harming the website no one will use the website again. Now I've never owned a website or had to deal with finding advertising companies support my website but there are more than enough choices to find a non-aggressive ad. I never really thought about looking at ads and ad-blocking from a Christian standpoint but even after thinking about it, I don't feel bad about using ad-block.

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Data Structure Ethic Paper: “Accidents Happen Autonomously?”

Taylor Perry

Dr. Paul West

Data Structures Analysis

11/30/18

Accidents Happen Autonomously?

When you think of AI what is the first thing that comes to your mind? Your smart phone, maybe if you’ve heard of a supercomputer or possibly some little household vacuum that goes around by itself learning where to go and where not to go. If something like that came to your mind then, good, it seems that you know about the friendly AI that we use to help us, but what about the AI that we use to protect us. It could be from a program protecting future cyber attacks, or maybe the “Eye in the Sky” drones, or even the very rockets we use to make sure targets aren’t a threat anymore. (Joshi, 2018) Whatever you thought of, either friendly or threatening, the question with AI is not what it can do but what should we apply it to.

When the military uses their drones for surveillance and are given the option of taking out a target with a missile and the AI in the drone is telling its code that the target is a enemy bunker. Turns out that the enemy bunker is a school and instead of terrorists you now have a few hundred deceased children and teachers. If a manual operator was in charge and saw this school and decided to use a missile to take out the target there would be serious actions taken against that human pilot of the drone. What do you do though if there is no pilot and this error is due to the drones AI? The blame wouldn’t go to the military who used the product but the manufacture. Gary Lea mentions in his article that its not the farmer who gave proper commands to the AI tractor, but the one who produced/provided that product. (Lea, 2015) This way there wont be multiple groups who use the AI product to become infamous but the actual inventor of the product. So the military should be the

first to blame but the company that told them that this product works perfectly and will get the job done nice and easy. The thing with AI is that it is always learning, either in simulations or real world tests, but it is always learning. In war though it is always changing and if AI is copying that has already happening and learning from that then it will always be a step behind new attacks. Not saying that it shouldn't be utilized, if it were to learn an attack once then it could use that to stop it in the future if it can see a similar pattern of attack. So for the drone not to hit the school it needs to know that a school is located at these locations and not to fire upon this location. The down side to this is that the enemy will learn that a school will not get hit and most likely use a school as a base.

My option though if AI has been a good or bad impact on society is that it has its ups and downs. I believe that is better protecting use on the battle field, sadly we don't hear all the stories of things gone wrong, but its also a good thing in our growing society as well. Unfortunately, we are not where we want to be yet to replacing some human workers for autonomous workers. For example the Tesla Company is using AI computer arms to assemble cars. (Hawkins, 2018) What seemed like a good idea turned out to not be the best decision, for now.

In conclusion with all things regarding to autonomous programing it takes time and testing to make a product that will eventually have a small error rate. Being a drone in a war or a car to make long trips a care free experience, but keep in mind the best way for a AI product to learn is to mess up. And ask yourself , what if you were the one in that mess up?

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Data Mining Review and Analysis: "A Genetic Algorithm-Based Feature Selection"

Taylor Perry

Data Mining

Dr. Babatunde

11/13/2018

Review and Analysis: A Genetic Algorithm-Based Feature Selection

By Taylor Perry

This article that was researched by Leisa Armstrong, Leng Jinsong, and (the one and only) Oluleye Babatunde is about the use and application of a genetic algorithm used in specifically identifying different leaves and the distinguishing between different plants based on this algorithm. In this paper you will find out how the Genetic Algorithm being used is developed and implemented using data to get certain results that users will be looking for. There are also different methods used to help better the search results and make the genetic algorithm work at a higher accuracy. Finally in the end I will discuss what this paper is good at, and where it can improve.

A genetic algorithm is an adaptive heuristic(enabling a person to discover or learn something for themselves) search algorithm that belongs to the larger part of evolutionary algorithms. (Genetic Algorithms, n.d.) This specific search is used when evolution or mutations are involved and even more so when it has to deal with evolution. Since this research is dealing with evolution and discrimination of different plant leaves it deals with many terms that would be in genetics that can correspond to genetic algorithm terminology. For this

Genetic Algorithm there are five certain points that are being looked at: chromosome encoding, fitness evaluation, selection mechanisms, genetic operations, and criteria to stop the genetic algorithm. There are also certain operations that are happening in the genetic algorithm that are iterative procedures that manipulate one or more populations of the candidates to make a new population with crossover and shows support for the principle of reproduction and survival of the fittest. Outside of this paper a genetic algorithm can be also used in different problems such as with: Robotics, optimized telecommunications routing, encryption and code breaking, and with the biomimetic invention that is used in this paper.

As for the methods used the first one that I came across is the CrossOverKid where it took two different chromosomes and then would scan both the p1 and the p2 which were different parent chromosomes and it would read the numbers from left to right and then combine them into making a child. The random number in the gene is assigned a position and when the position of the number value is less than 1 than the gene at the position 1 is flipped. This is resulting in the Least Significant Bit and the Most Significant Bit of the child chromosome. Some other methods and function used are the FeatIndex which is used to get the index from a binary chromosome, and NewDataset which takes the old data and runs the parents to make children DNA.

Weakness of the paper is that it seems to general and is more for those who are already juniors or seniors in computer science which is good, but the interest of the paper for does nothing to attract the beginning levels of computer science or AI learning. If I remember your opening slideshow about you presented this as machine learning and something about biology. Its good too see it used in your class about data mining and how it applied with the process of obtaining data and what you do with it. It would have helped if we made the data at first, making it small but what was even better was that we had to used data from the internet or one that was already in the library to make charts on. Sadly I cant say to much more about the paper since my knowledge

and understanding of botany and machine learning are limited I can say that it is interesting to see the application and building of how an application can distinguish between different flowers and continues to grow with each submission of entries.

Presentation(s)

Senior Project: N/a

Computer Architecture : Sign Extender

Grade 1/1

Feedback: N/a

SIGN EXTENDER

BY TAYLOR PERRY(PYROCAMPER)

WHAT IS IT??

- The process of increasing bits.
- Most commonly returns a value of 32 bits.
- If the first value is a 0, then it will add on more 0's keeping it positive
- If the first value is 1, then it will add 1's to keep the number negative.

PROBLEMS I HAD

- If you give it the wrong value(16 bits) it will output the wrong.
- This is structural flow
- The delay is 0
 - There are no gates therefor no gate delay

THE END